DO NOW — RECALL

Lesson 069 · Data Mining and Visualisation of Patterns

Name: _____ Class: _____ Date: _____

1. What is a backtracking branch?

2. What is an outlier?

3. What is the purpose of a scatter graph?

4. Array [2,3,4,5]: double elements at indexes 1 and 3. Give updated array and sum.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

One choice explored by search.

2. Model answer

Value unusually different from others.

3. Model answer

Displays association between two numerical variables.

4. Model answer

[2,6,4,10]; total22.

By the end, you can…

• Data mining seeks patterns or associations across sets of data; define the question, prepare records and examine the results.

• Visualisation makes patterns and anomalies visible. Scatter charts suit pairs of numeric data; bar charts compare categories.

• A correlation does not establish cause. Sampling, confounders, missing values and measurement choices may mislead interpretation.

Vocabulary — know these terms

data mining — Finding patterns in datasets.

scatter plot — Pairs of numeric variables plotted as points.

outlier — Unusually distant observation.

correlation — Association, not proof of causation.

NEW LEARNING

Read, annotate and explain

Define and predict

Data mining seeks patterns or associations across sets of data; define the question, prepare records and examine the results.

Explain

Visualisation makes patterns and anomalies visible. Scatter charts suit pairs of numeric data; bar charts compare categories.

Evaluate

A correlation does not establish cause. Sampling, confounders, missing values and measurement choices may mislead interpretation.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

For points (1,2),(2,4),(3,6), state trend and one limitation.

Follow the reasoning

Separate observation from causal explanation.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

For points (1,2),(2,4),(3,6), choose a suitable chart and describe association. [4]

Question 2

For [40,42,46,96] calculate mean and median with working. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Evaluate mining a school's student-usage logs to predict attainment. Address fairness, data quality and privacy. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A rising scatter trend proves that extra practice alone caused higher marks.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.